

Cover Sheet

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Keywords:

Synthetic Data, Counterfactual Fairness, Causal Fairness, Generative Models, Health Equity

Potential Mentors:

Drago Plecko, Emma Pierson, Emily Alsentzer, Iren Y Chen, Elias Barenboim, Russ Altman, van der Schaar

Past Doctoral Consortium Participation:

7th Annual Conference on Health, Inference, and Learning (CHIL, 2025, Berkeley, CA)

Planned Doctoral Consortia/Conference Attendance in 2025-2026:

Stanford Causal Science Center Conference, Nov 2025 (Abstract in Preparation)

Machine Learning for Health Symposium, Dec 2025 (Demo in Preparation)

Pacific Biocomputing Symposium, Jan 2026 (Abstract in Preparation)

Balancing Fairness and Utility Constraints for Synthetic Data Generation

Nitish Nagesh

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Abstract

Synthetic data generation creates data based on real-world data using generative models. In health applications, generating high-quality data while maintaining fairness for sensitive attributes is essential for equitable outcomes. Existing fairness-aware synthetic data generation models focus on satisfying counterfactual fairness constraints. There is a need to go beyond counterfactual fairness towards causal fairness to retain causal structure during data generation and evaluation. To address this gap, I led the development of two synthetic data generation pipelines - (i) "FairCauseSyn", an LLM-based tabular data generator that satisfies causal fairness for health settings (published in IEEE EMBC 2025) and (ii) "FairTabGen", a framework that takes into consideration both counterfactual and causal fairness for a wide range of data (submitted to AAAI 2026). My plan is to continue working on enhancing the framework by incorporating effects of unobserved confounders and improving evaluation on metrics such as benefit fairness and process fairness targeting potential submissions in NeurIPS or AAAI in Spring/Summer 2026.

Introduction

Synthetic data has become an essential tool in machine learning, offering a way to mitigate privacy risks, address data scarcity, and promote equity in high-stakes domains such as healthcare, criminal justice, and education (Jordon et al. 2022). Recent advances in generative modeling such as generative adversarial networks (GANs), variational autoencoders (VAEs), diffusion models, and large language models (LLMs) have made it increasingly feasible to generate realistic tabular datasets. However, while models like CTGAN (Xu et al. 2019), TabDDPM (Kotelnikov et al. 2023), REalTabFormer (Solatorio and Dupriez 2023) and BeGReaT (Borisov et al. 2022) emphasize statistical fidelity, they often fall short in addressing fairness, especially when structural biases are encoded in the training data.

Related Work

Despite advancements in synthetic data generation, fairness remains a critical challenge. While counterfactual fairness has gained popularity as a formal notion of non-

discrimination, Schröder, Frauen, and Feuerriegel (2023) argue that this notion is insufficient for practical fairness evaluation due to three structural issues with respect to admissibility, ancestral closure and identifiability. To ameliorate these concerns, another paradigm of causal fairness (Plečko and Bareinboim 2024) has emerged. However, existing fairness-aware synthetic data generation methods such as FairGAN (Xu et al. 2018), DECAF (Van Breugel et al. 2021) primarily focus on satisfying counterfactual fairness constraints (Kusner et al. 2017).

Research Gap

Existing synthetic data generation methods either focus on a single notion of fairness (either counterfactual or causal fairness) and are not geared towards health settings. As illustrated in Fig. 1, there is limited research on exploring examining fairness, bias and utility trade-offs in synthetic data generation from a causal lens for health applications.

Research Question

In my thesis, I aim to answer the following research question:

How can we generate synthetic data that optimizes for causal fairness, counterfactual fairness while retaining data utility and fidelity?

Results as on Sep 30, 2025

I led the development of **FairCauseSyn** (Nagesh, Wang, and Rahmani 2025), an LLM-augmented data generation approach that aims to enhance causal fairness. The experiments on a real-world clinical data shows that the proposed method generates data with less than 10% deviation on fairness metrics. When trained on machine learning predictors, synthetic data reduces bias on sensitive attributes by 70% compared to models trained on real data.

Going beyond causal fairness, I built **FairTabGen** (Nagesh et al. 2025), an LLM-based synthetic data generation framework that incorporates both causal fairness and counterfactual fairness in the data generation and evaluation phase. Across datasets in the law school admissions, COMPAS and MIMIC, the method outperforms state-of-the-art by 10% on fairness-related metrics while retaining statistical utility

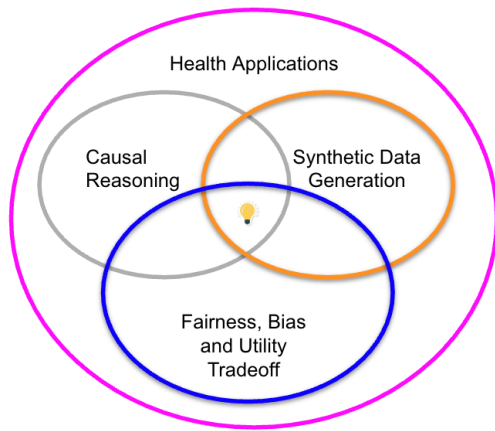


Figure 1: Research Themes and Intersectionality

while using less than 20% of original data, demonstrating efficiency in low-data regimes.

Planned contributions by Jan 20, 2026

I am working towards revising my submission about balancing counterfactual fairness and causal fairness along with utility constraints from AAI. The components to add would be more related baselines, enhance the synthetic data generation algorithm and perform extensive experiments for health-related case studies. I plan to submit the revised work to the International Conference on Machine Learning (ICML) by January 30, 2026.

Targeted submission by Aug 15, 2026

In the future, I aim to extend this work further into two strands. The first one would be to simulate the effect of unobserved confounders using the extended fairness map into the data generation process (Plečko and Bareinboim 2024; Schröder, Frauen, and Feuerriegel 2023). Another strand is to enhance the evaluation to analyze fairness from a decision maker perspective via benefit fairness (Plečko and Bareinboim 2024) to ensure synthetically generated data is beneficial for advancing health equity. I plan to submit this work either to NeurIPS in May 2026 or AAI in August 2026 depending on my results and progress by that time.

Resources and Mentorship

I will continue working with my advisor Prof. Amir Rahmani from University of California Irvine, who has expertise in multimodal conversational health agents, and my collaborator Dr. Randall Stafford from Stanford, who provides clinical expertise on fairness and bias in healthcare. I have applied for the NVIDIA fellowship and internal department-wide fellowships. I will leverage computational capabilities to run large scale models to generate data for electronic health record (EHR)-style datasets such as MIMIC-IV and proprietary observational health data to validate my approach.

Anticipated Thesis Contribution

The goal of my thesis is to develop and evaluate synthetic data generation techniques that take into consideration fairness and utility metrics both in the data generation and evaluation process. Doing so would allow the generated data to be used for downstream machine learning prediction tasks with limited bias. Since the application of my framework is mostly in health settings and generative models are on the rise, being cognizant of fairness in the entire machine learning pipeline will contribute to improving health equity.

References

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Nitish Nagesh

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EDUCATION

Ph.D. Computer Science, University of California Irvine, GPA: 3.90/4.0	Sep 2021 - Dec 2026
M.S. Computer Science, University of California Irvine, GPA: 3.83/4.0	Sep 2021 - June 2023
M.S. Computer Engineering, Technical University of Munich, Germany GPA: 3.73/4.0	Sep 2018 - Dec 2020
B.E. Electrical Engineering, R.V. College of Engineering, India, GPA: 3.65/4.0	Sep 2012 - June 2016

PROGRAMMING SKILLS AND TOOLS

- Python, SQL, Pandas, Tensorflow, SciPy, Snowflake, Git, AWS, Confluence, Jira, Figma, Canva, Trello

PROFESSIONAL EXPERIENCE

Graduate Research Scientist, UC Irvine Institute for Future Health, Irvine, CA June 2022 - Present

- Build novel LLM-augmented causally fair synthetic data generation pipeline using Python, R to reduce bias in real-world health datasets; first-author publication in IEEE Medicine and Biology conference.
- Assisted in conceptualizing an LLM-powered framework for a diet recommendation chatbot using health data, achieving 92% effectiveness; co-authored publication in Elsevier Smart Health Journal.
- Integrated dietary guidelines and nutrition calculation tools into LLM-based health agents, enabling explainable diet risk assessments; co-authored publication in IEEE Medicine and Biology conference.
- Developed a causal inference framework using Python for a 3-year N-of-1 observational dataset to analyze the effect of caffeine on heart rate variability; first-author paper in computing conference.
- Implemented an image segmentation pipeline for a retina dataset using deep learning frameworks such as TensorFlow, Keras, SciPy achieving 31% accuracy for optic disk and 8% for fovea segmentation.

Research Data Scientist Intern, ResMed, San Diego, CA June 2024 - September 2024

- Built LLM-powered topic modeling framework to analyze quantitative data for 6,000 sleep apnea patients using LLMs and Langchain in Python, improving therapy personalization.
- Collaborated with an interdisciplinary team of 5 researchers to refine product claims related to pressure and wakefulness in sleep apnea patients; first-author abstract in American Thoracic Society 2025.

CEO & Founder, FoodMoodyAlly, Irvine, CA January 2023 - December 2023

- Lead planning, coordination and development of AI-driven food recommendation systems to improve mood of pregnant women & new mothers leveraging Python, Xcode, Trello, Git & AWS.
- Created user personas in Figma based on 30+ customer interviews using NSF I-Corps grant (\$5,000).
- Winner Product Design Competition (1st out of 17 teams), Mental Health Hackathon \$3,500 (1 of 30).

Data Scientist Intern, iHealth Labs, Sunnyvale, CA June 2023 - September 2023

- Led de-identification, aggregation, and analysis of data for 12,000 patients in a remote patient monitoring program, adhering to HIPAA guidelines, resulting in an academia-industry partnership.
- Collaborated with a cross-functional team to develop personalized interventions for diabetes and hypertension management; co-authored abstract and poster at American Heart Association 2024.

Software Engineer, Qualcomm, Austin, TX March 2021 - August 2021

- Developed a Python tool to parse 5,000 logs from the Qualcomm AI accelerator, reducing cycle time by 3x and lowering upstream production costs.
- Triaged and debugged failures in three ML accelerator SDKs through feature engineering and model evaluation, resulting in a 10% performance improvement across internal benchmarks.

Visiting Researcher, University of California San Diego, San Diego, CA September 2019 - February 2021

- Developed novel reliability-aware task allocation strategies for IoT networks using Python to reduce overall maintenance cost.
- Built a real-world IoT mesh network communicating via MQTT and Wi-Fi to measure impact of resource constraints on reliability leading to 3 co-authored publications in top-tier system conferences.

RELEVANT PUBLICATIONS

- [Submitted to AAAI 26] Nitish Nagesh*, Shakibhamedan* et al. "FairTabGen: Unifying Counterfactual and Causal Fairness in Synthetic Tabular Data Generation." * equal contribution
- Nitish Nagesh et al. "FairCauseSyn: Towards Causally Fair LLM-augmented Synthetic Data Generation.", IEEE Medicine and Biology Conference, Copenhagen, 2025.
- [Abstract] Nitish Nagesh et al. "Assessment Of AI/ML Approaches For Qualitative Analysis In Obstructive Sleep Apnea", American Thoracic Society, San Francisco, 2025.
- [Abstract] Yong Huang, Ajan Subramanian, Nitish Nagesh et al. "Patient Adherence to Self-Monitoring Practices and Glycemic Control - Findings from a Multiyear Digital Coaching Program", American Diabetes Association, 2025.
- [Under Review] Ajan Subramanian, Nitish Nagesh et al. "Remote Blood-Pressure Monitoring Improves Control in Stage II Hypertension", Journal of the American Heart Association, 2025
- Mahyar Abbasian, Nitish Nagesh et al. "Knowledge-Infused LLM-Powered Conversational Health Agent: A Case Study for Diabetes Patients", IEEE Medicine and Biology Conference, 2024.
- Zhongqi Yang, Nitish Nagesh et al. "ChatDiet: Empowering personalized nutrition-oriented food recommender chatbots through an LLM-augmented framework.", Elsevier Smart Health, 2024.
- [Abstract] Ajan Subramanian, Nitish Nagesh et al. "Long Term Remote Patient Monitoring Reduces Blood Pressure in Patients with Stage II Hypertension.", American Heart Association, 2024.
- Nitish Nagesh, Iman Azimi et al. "Towards Building Deep Personal Lifestyle Models using Multimodal N-of-1 Data.", 29th International Conference on Multimedia Modeling, Norway, 2023
- Ali Rostami, Nitish Nagesh, et al. "World food atlas for food navigation." Proceedings of the 7th International Workshop on Multimedia Assisted Dietary Management, Portugal, 2022.

LEADERSHIP AND SERVICE

Teaching Assistant, UC Irvine

Sep 2021 - Present

- Mentored 600+ students across writing, programming, data structures, and language courses, supporting instruction in collaboration with 5+ faculty.
- Designed and facilitated coding, design, and documentation support across multiple quarters, earning >90% student satisfaction in engagement and clarity.

Lead Mentor, Artificial Intelligence Club, UC Irvine

Sep 2021 - Present

- Initiated "How to apply to Grad School" series for 75+ students and reviewed 15+ personal statements leading to 5+ admission offers from CMU, UC Berkeley, etc.

DEI Representative and Peer Mentor, CS Department, UC Irvine

Sep 2021 - Present

- Ally for women in tech (GHC '23 scholar), peer mentor for 5 underrepresented students, led panel of 50+ attendees on ethical and societal challenges in computing.

Volunteer at Non-Profit, SMVA Trust, India

July 2016 - Aug 2018

- Led the "Feeding the Hungry" project delivering food and essentials to 10+ orphanages and senior centers to alleviate poverty and hunger.

HONORS AND AWARDS

- **UC Irvine** - Entrepreneur Award in Computer Science \$5,000 (1 per year), Fellowship to commercialize research \$2,500 (1 per year), Fellowship to coach student entrepreneurs \$5,000
- **TU Munich, Germany** - International Student Scholarship US\$2,500 (1 of 200) for academic excellence, Masters Thesis Scholarship US\$2,000 (1 of 50), Young Academy Scholar Cohort (1 of 40)
- **R V College, India** - Best Outgoing Student (1st in 70) for all round excellence, Academic Excellence Award (2nd in 70) for highest overall GPA, Most Innovative Thesis Award (2nd in 70)

INVITED TALKS

- LLMs/AI tools on a Daily Basis, Pacific Club, Feb. 2025, Newport Beach, CA
- Building Diet and Well-being Recommendation Systems for Lifestyle Management
 - Stanford University, Oct. 2023; IIIT-Delhi, India, Dec. 2022; INRAE, Paris, Oct. 2022; University of Barcelona, Spain, Oct. 2022

Personal Statement

Nitish Nagesh, Ph.D. Student in Computer Science, University of California Irvine

I am excited to attend the doctoral symposium at AAAI 2026 to add value by sharing my diverse experiences in academia, industry and entrepreneurship. I am eager to learn from experts in the field of causal reasoning, synthetic data and health equity.

Slated Contributions to the Doctoral Symposium:

I look forward to sharing my recent research on fairness-aware synthetic data generation that can lead to broader conversations and engagements about health equity, AI safety and aspects of leveraging AI for social good. Going deeper in research, I am excited to share my experience in developing causal models for large-scale time series observational studies for remote patient monitoring. Beyond research, I am keen on sharing experience working as a data scientist intern at health tech companies in the U.S. and my entrepreneurial experience in building a startup (\$10,000 USD funding) geared towards improving mental wellbeing of new mothers through timely dietary interventions.

Leveraging Conference Participation and Presentation Opportunity:

At UC Irvine, I am a mentor for the Artificial Intelligence Club, a Grace Hopper Scholar awardee (i.e. ally for women in tech), and computer science representative for broadening community participation. Being in-person at the conference will help me motivate other students, especially from underrepresented backgrounds to pursue AI research thereby bridging the information gap and increasing diversity in conference participation. I hope to be a better mentor for undergraduates and help them find research opportunities off-campus. I motivated my sister to apply to the undergraduate consortium at AAAI as a freshman in India. During my conference at ACM multimedia, I was awarded the best social media reporter. I aim to bring the same level of enthusiasm to the venue by attending poster sessions, lightning talks and cheering my fellow doctoral students.

Plan of Action:

Before the conference, I plan to reach out to authors/mentors via email initiating the conversation and exploring people I want to interact with. During the conference, I will meet with them for coffee, attend their talks/sessions, take pictures and connect with them on LinkedIn. Post conference, I will make a LinkedIn post sharing my experience, followup with people I met and explore potential avenues for synergy in the future.